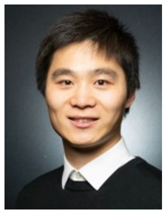
Dr. Jimmy (Huanming) CHEN –PhD CEng CSci FRSA FHEA



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University of Oxford, Department of Engineering Science, Oxford, OX1 3PJ.

Nationality: British. Portfolio: LinkedIn; Google Scholar; ORCID

Dr. Jimmy H. Chen is an academic leader and professional engineer specializing in advanced materials science, solid mechanics, and innovative engineering education. As a Chartered Engineer and Chartered Scientist with extensive pedagogical expertise, he champions a practice-based, inclusive approach to teaching grounded in experiential learning and Universal Design for Learning. Beyond the classroom, Dr. Chen maintains an active research focus on lightweight, energy-absorbing, and bio-inspired systems, bridging structural optimization with advanced manufacturing to develop next-generation resilient applications while engaging widely through public scholarship and industry partnerships.

Education

D.Phil (oxon)	Solid Mechanics, Dept. of Engineering Science, University of Oxford, UK *Thermomechanical behaviours of thermoplastic TPU - DYMAT Thesis Awarded*	2017 - 2021
PGCert (Level 7)	Postgraduate Certificate in Academic Practice, Queen Mary University, UK	2025 - 2026
B.Eng. (1st Class Hons)	Civil Engineering, University of Nottingham, UK - *International Scholarship*	2013 - 2016
Accredited Certificate	Advancing Teaching and Learning Programme, University of Oxford, UK	2025
	IOM3 Certificate in Management: *Leadership, Performance, Finance & Risk, Project Management*, Institute of Materials, Minerals and Mining (IOM3), UK	2025

Professional Experience

Assistant Professor in Mechanical Engineering	Teaching, Leading, Curriculum Design (BEng/MEng), Mechanical Engineering, New Model Institute for Technology and Engineering (NMITE), Hereford, UK. Teaching Modules List: *Year 1: Statics and Dynamic pf Complex Mechanics* *Year 2: Mechanics of Materials* *Year 3: Research and Modelling*	2026 - Now
Teaching Associate in Mechanical Engineering	Teaching, Leading, Curriculum Design (BEng/MEng), Mechanical Engineering & Materials Science, Queen Mary University of London, UK. Teaching Modules List: *Year 1: Mathematics for Engineering, Sustainable Materials (solar science)* *Year 2: Applied Solid Mechanics, Heat and Mass Transfer* *Year 3: Materials and Nonlinear Structures, Engineering Drawing and Design* *Year 4: Sustainable Engineering Management, Finite-Element Modelling*	2025 - 2026
Research Fellow & Visiting Research Associate	Department of Engineering Science, University of Oxford *Design, characterisation, and modelling of machine-learning-driven structure with functionality in robotics dynamics.*	2025 - 2026
Postdoc-research Associate	Department of Engineering Science, University of Oxford, UK *UKRI: Experimental mechanics, chemo-mechanics and modelling, material processing in polymers and composites; collaborated with Prof. Laurence Brassart.*	2022 - 2025
	USAF: Dynamics of materials, impact mechanics and modelling, polymer processing and experimental mechanics in polymer; collaborated with Prof. Clive	2021 - 2022

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Chartered Status	Chartered Engineer (CEng) & Chartered Scientist (CSci), IOM3, UK	2025
	Chartered Physicist (CPhys), IOP, UK	2026
Fellowship / Affiliation	Fellow of Royal Society of Arts (FRSA)	2026
	Fellow of the Higher Education Academy (FHEA)	2026
	IOM3 Accreditation & Approval Assessor (for Registered/Incorporated and Chartered grades)	2026 – Now
	Programme Lecturer of Worcester College; Tutor of Wolfson College; Spotlight TEDx Speaker of Oxford University; Keynote Speaker of Wolfson Research Day	2024 – Now
Member of Professional Body	IOM3 (Institute of Materials, Minerals & Mining, UK); IOP (Institute of Physics, UK); CWMS (Coventry & Warwickshire Materials Society); BSSM (British Society for Strain Measurement, UK); SEM (Society for Experimental Mechanics, USA); oSTEM (Supporting LGBT+ in STEM Research).	2019 - Now
Funding	Industry Funding (*£120k*), Fuerda Smartech Co., Ltd., Ingolstadt, Germany; Education Scholarship (*£1k*) in Policy and Examine Pedagogy, QMUL.	2025 2026

Teaching Experience (modulus)

UG Year 1 – Sustainability (Heat Exchanger Design, Solar Energy, Photovoltaic) (186 registration) 2026
Undergraduate lecturer for Mechanical Engineering; co-taught with Prof. Huasheng Wang & Dr. Flurin Eisner

UG Year 2 – Applied Solid Mechanics & Lab-class (248 registration) 2025 - Now
Undergraduate lecturer for Mechanical Engineering; co-taught with Dr. Tao Liu.

UG Year 2 – Heat and Mass Transfer & Lab-class (220 registration) 2025 - Now
Undergraduate lecturer for Mechanical Engineering; co-taught with Dr. Hicham Adjali.

UG Year 1 –Essential Mathematics Skills for Engineers & Tutorial (370 registration) 2025
Undergraduate lecturer for Mechanical Engineering, Bio-engineering Engineering & Chemical Engineering

PG/UG Year 4 –Engineering Design Optimisation and Decision Making (116 registration) 2025
Postgraduate lecturer for Mechanical Engineering; co-taught with Dr. Jun Chen.

UG Year 2 - Project Management on Structure, Design and Construction, (25+ registration) 2025
Engineering teacher and tutor on Oxford Certificate Programme at Worcester College.

UG Year 3 - Polymer, Composites, Ceramic (30+ registration) 2024 - 2025
Engineering teacher/keynote speaker in Wolfson College Engineering Society; co-taught undergraduate students from engineering, chemistry, physics, and medical science departments.

UG Year 3 - Materials, Metals & Lab-class (45+ registration) 2024 - 2025
Undergraduate lecturer for Mechanical Engineering Course; co-taught with Prof. Reece Oosterbeek.

UG Year 1 - Structures, Mechanics & Lab-class (90+ registration) 2024
Undergraduate lecturer for Engineering Science Course (MEng); co-taught with Dr Branoc Richards.

UG/PG Year 4 – Material Testing Lab-class (90+ registration, lab manager) 2019 - 2025
Managing, training, organizing, supervising PhDs, demonstrating and teaching MEng students on 4 Labs, including Material Testing Lab (tension/compression, bending, torsion, shear, programming on MTS/Instron), Thermal Mechanical Lab (DSC, DMA, TGA, Rheometer), Impact Engineering Lab (high-rate SHPB/ballistic impact test), 3D Printing Lab.

Journal Articles (Google Scholar: Huanming CHEN Oxford; ORCID: 0000-0001-9864-1890)

Structure and Mechanics

- Zhang, Y., ..., **Chen, H.***, 2026. Mechanical characterisation and prediction of thermal evolution in horseshoe microstructure lattices under soft impact. *International Journal of Solids and Structures*. 339 (2026): 114134. 2026

Liu, W., ..., **Chen, H.***, 2026. Anisotropic plasticity and ductile fracture of magnesium alloy under biaxial loading: 2026

Experiments and modelling. *Engineering Fracture Mechanics*. p.112278.

Liu, W., ..., **Chen, H.***, 2026. Stress-state and strain-rate dependency of fracture in SCFR-PEEK composites under biaxial loading. *Composites Part A: Applied Science and Manufacturing*. p. 109594 • 2026

Long, S., **Chen, H.**, ..., 2025. Response of shear thickening fluids to high velocity ballistic impact. *International Journal of Impact Engineering*: 105248. 2025

Zhang, Y., ..., **Chen, H.***, 2025. Nonlinear mechanics of horseshoe microstructure-based lattice design. *International Journal of Mechanical Sciences*, 285, p.109781. 2025

Chen, H., et al., 2022. Experimental characterisation and modelling of the strain rate dependent mechanical response of a filled thermo-reversible supramolecular polyurethane. *International Journal of Impact Engineering*, 166, p.104239. 2022

Chen, H., et al., 2020. Application of linear viscoelastic continuum damage theory to the low and high strain rate response of thermoplastic polyurethane. *Experimental Mechanics*, 60, pp.925-936. 2020

Material Science

Pan, Z., **Chen, H.**, ..., 2026. Constitutive modelling of hydrolytic degradation and viscoplasticity in glassy polymers: An effective temperature approach. *International Journal of Plasticity*. 104690. 2026

Chen, H., et al., 2025. Hydrolytic degradation of amorphous PLA: effect of mechanical loads. *Polymer Degradation and Stability*, 111751. 2025

Ferreira, T., **Chen, H.**, ..., 2025. Hydrolytic degradation behaviour of electrospun filaments for biological tissue repair. *Journal of the Mechanical Behavior of Biomedical Materials.* 175: 107308. 2025

Pan, Z., **Chen, H.**, ..., 2024. Constitutive modelling of glassy polymers considering shear plasticity and craze yielding. *International Journal of Plasticity*, 178, p.103996. 2024

Chen, H., et al., 2023. Shear yielding and crazing in dry and wet amorphous PLA at body temperature. *Polymer*, 289, p.126477. 2023

Chen, H., et al., 2022. Characterization and modelling of bisphenol-a type epoxy polymer over a wide range of rates and temperatures. *Polymer*, 249, p.124860. 2022

Chen, H., et al., 2021. Mechanical characterisation and modelling of a thermoreversible supramolecular polyurethane over a wide range of rates. *Polymer*, 221, p.123607. 2021

Chen, H., 2021. Thermomechanical behaviours of unfilled/filled thermoplastic polyurethane. Diss. University of Oxford. Awarded to *DYMAT Publication Collection 2021*. 2021

CPD - Honors and Awards

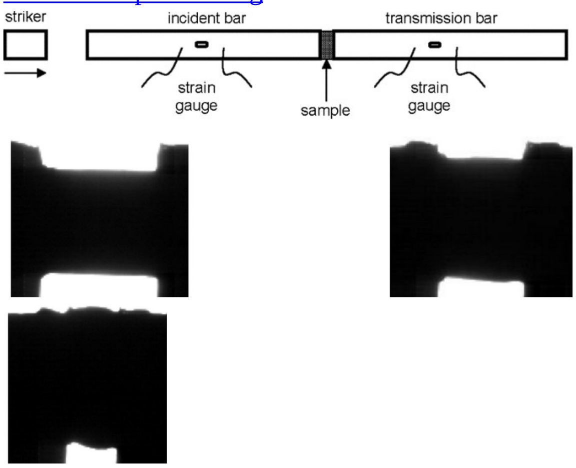
Wolfson Interdisciplinary Award	3-years continuous contributions to Wolfson Interdisciplinary Engineering Day	2026
Spotlight: Illuminating Engineering	Selected projects, showcasing ground-breaking research that improves our lives and expands our horizons, highlighted University of Oxford research achievements and were presented in a *TEDx Talk* organised during the Oxford City and University Summer Research Fair at Begbroke Campus.	2025
Entrepreneurship Award (for Final Project Selection)	*Pitch@Palace 12.0*, London, UK. - For the outstanding entrepreneurship project have been selected, awarded, and presented in the St. James Palace, London, UK.	2020
Entrepreneurship Award (for Best Outstanding Research)	*Bioescalator*, University of Oxford, UK. - For the outstanding Bio-engineering project have been selected, awarded, and presented.	2019
Entrepreneurship Award (for Best Potential Investment)	*Venture Fest*, University of Oxford, UK. - For the outstanding Bio-engineering project have been selected, awarded, and presented.	2019
Entrepreneurship Award (for Best Biology Project)	*Biomaker Challange*, University of Cambridge, UK. - For the outstanding Bio-engineering project and bio-sensor design have been selected, awarded, and presented.	2018

Project Highlight (Flyer)

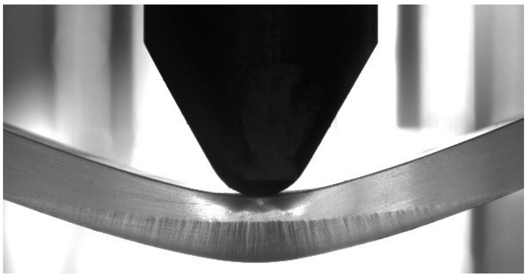
Keywords: *Polymers-Composites*; *Advanced Manufacturing*; *Experimental Mechanics*; *Multi-Scale Modelling and Simulation*; *Structural Design and Optimisation, Chemo-Mechanical Characterisation*.

Research Highlights

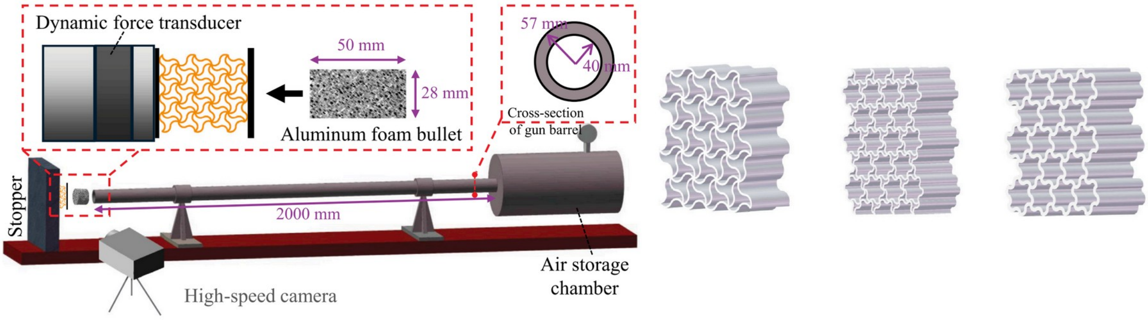
Material impact testing



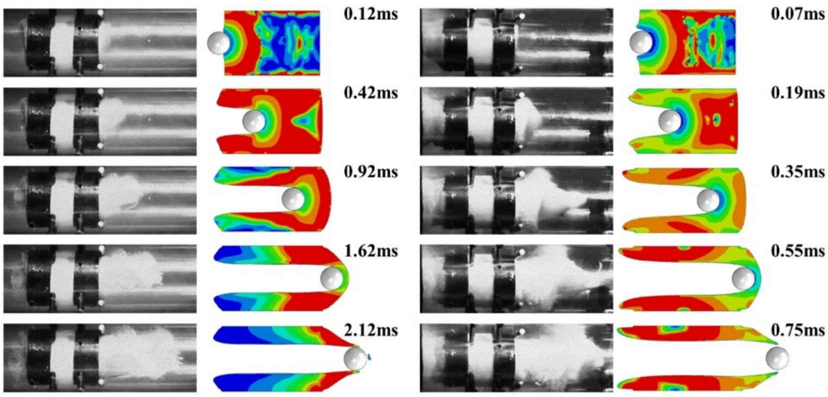
Material damage and failure



Horseshoe microstructure lattices

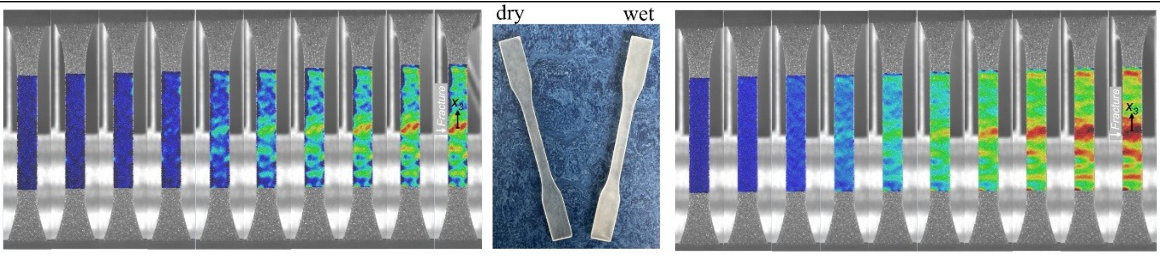


Ballistic resistance testing

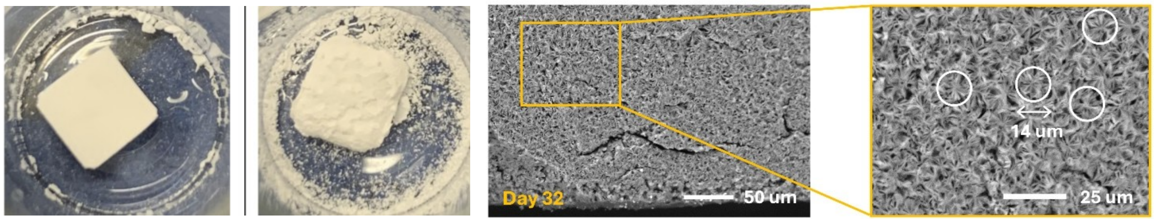


TED Talk - "Spotlight Engineering", organised by University of Oxford & Youtube Link

Evaluating damage with digital image correlation techique



Hydrolytic degradation mechanism at biodegradable polymer



Self-healable (full-recovery) polymer at body temperature